

Cluster Analysis & Visualisation

Live Q&A – 12 March 2026

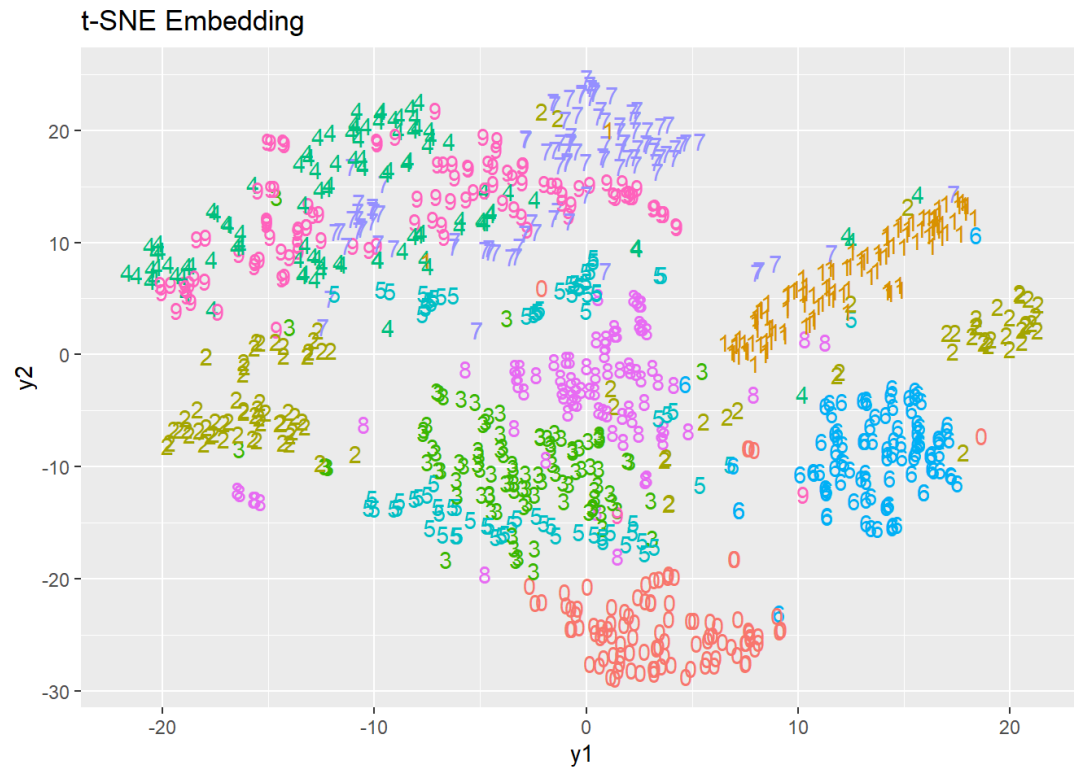
Niall Adams

Overview

- Coursework 2
- Guest speaker: Dr Joshua Neil, Co-founder of Alpha Level
- (Time allowing): Types of clustering - k -means, hierarchical
- Clustering as an exploratory tool

- Q&A

Clustering



k-means Clustering

- Clusters: 'grouping of observations where within-group variation is less than between-group variation'

- Aim is...

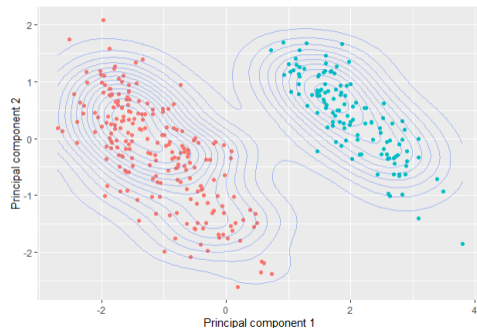
- given the number of clusters present in the data
- to find the allocation of observations to clusters that minimises the within-cluster variation

within-cluster sum of squares (WCSS)

$$= \sum_{i=1}^K \sum_{j=1}^{n_i} \|x_j - \mu_i\|^2$$

- Implementation in R:

```
penguin_pca_km2 <- kmeans(penguin_pca$x[,1:2], centers=2)
```

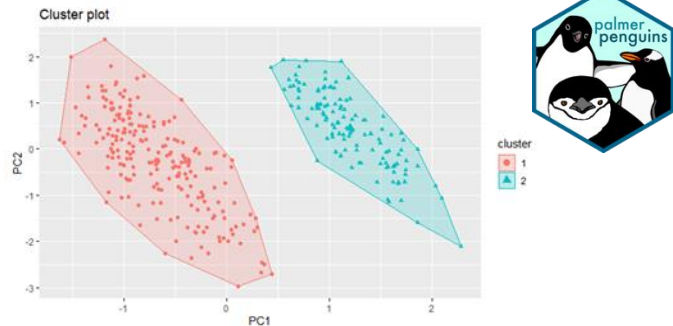


k-means Clustering

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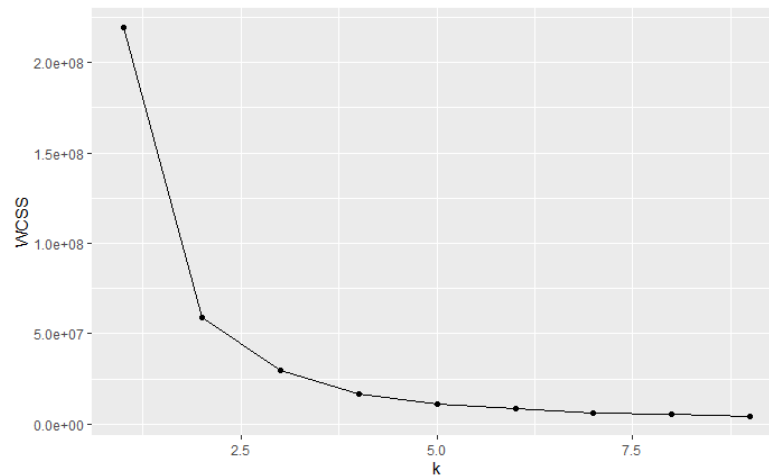
- Implementation in R:

```
penguin_pca_km2 <- kmeans(penguin_pca$x[,1:2], centers=2)  
fviz_cluster(penguin_pca_km2, data = penguin_pca$x[,1:2])
```



Clustering as an Exploratory Tool

- In EDA, some questions of interest:
 - How many clusters?
 - Does dimension reduction help clustering?



Any Questions?
